

Os skill 13

13(a) Input Redirection

What it does: Reads input from a file by redirecting stdin using dup2().

```
GNU nano 8.7.1 skill13a.c *
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("input.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("input.txt");
        return 1;
    }

    /* Redirect stdin to file */
    if (dup2(fd, STDIN_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    printf("Reading from input.txt:\n");

    while (fgets(buffer, sizeof(buffer), stdin) != NULL)
    {
        printf("%s", buffer);
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano skill13a.c
sri_dhruthi@Dhruthi:~$ gcc skill13a.c -o skill13a
sri_dhruthi@Dhruthi:~$ ./skill13a
Reading from input.txt:
Operating System Lab
File Copy using System Calls
```

13(b) Output Redirection

What it does: Redirects stdout to an output file so that program output is stored in the file.

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 skill13b.c *  
#include <stdio.h>  
#include <stdlib.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main()  
{  
    int fd;  
  
    /* Create output file */  
    fd = open("output.txt",  
             O_WRONLY | O_CREAT | O_TRUNC,  
             0644);  
  
    if (fd == -1)  
    {  
        perror("output.txt");  
        return 1;  
    }  
  
    /* Redirect stdout to file */  
    if (dup2(fd, STDOUT_FILENO) == -1)  
    {  
        perror("dup2");  
        close(fd);  
        return 1;  
    }  
  
    close(fd);  
  
    printf("Hello from output redirection.\n");  
    printf("This data is written into output.txt\n");  
  
    return 0;  
}
```

```
sri_dhruthi@Dhruthi:~$ nano skill13b.c  
sri_dhruthi@Dhruthi:~$ gcc skill13b.c -o skill13b  
sri_dhruthi@Dhruthi:~$ ./skill13b  
sri_dhruthi@Dhruthi:~$ cat output.txt  
Hello from output redirection.  
This data is written into output.txt  
sri_dhruthi@Dhruthi:~$ ls -l output.txt  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 68 Sep 28 15:07 output.txt  
sri_dhruthi@Dhruthi:~$
```